CRESCENT STAR MARINE

Main Engine PLC & HMI Alarm Monitoring System

Budgetary Quotation — For Information Only

QUOTATION NO	CSM-ME-PLC-2026-0928	DATE	28 Sep 2026
CUSTOMER	.....	VESSEL / IMO	.....
PREPARED BY	Şükrü Umut Sanlı — Technical Department Director	CURRENCY	USD (US Dollar)

IMPORTANT NOTICE

This document is a **budgetary quotation for information purposes only and is not a final offer.** The price range stated will vary **according to your requirements** and the final system scope. The final price will be submitted in a formal quotation after an on-board survey and scope confirmation.

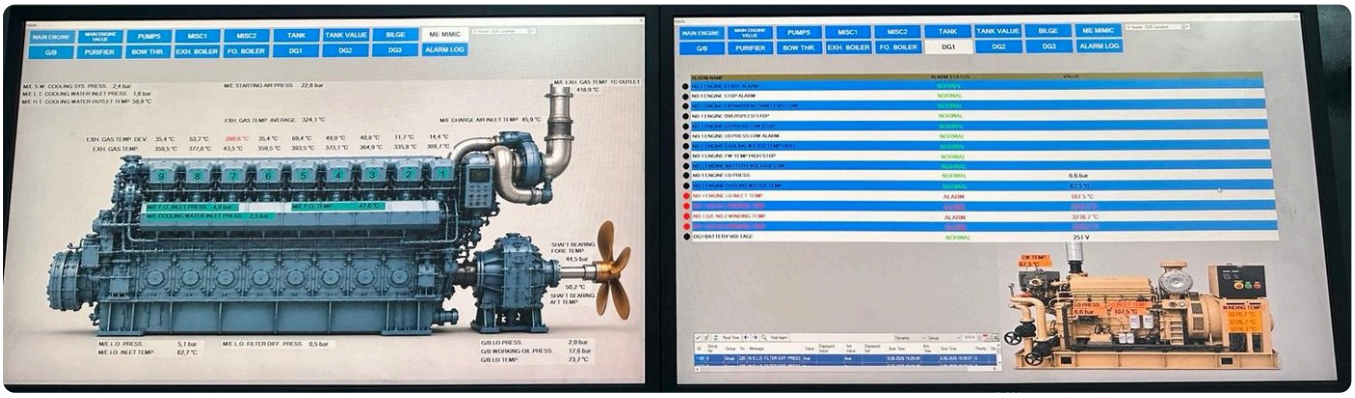
Dear Sir / Madam,

Please find below our budgetary pricing for the design, cabinet manufacturing, software and commissioning of a PLC-based alarm, monitoring and control system (PLC + HMI) for your vessel's main engine. The proposed system is built on Siemens SIMATIC PLC hardware, touch-screen HMI panels in the engine control room (ECR) and a dual-monitor SCADA workstation.

ESTIMATED BUDGET
USD 45,000 – 75,000
excluding VAT

PLATFORM
Siemens SIMATIC
PLC + touch HMI

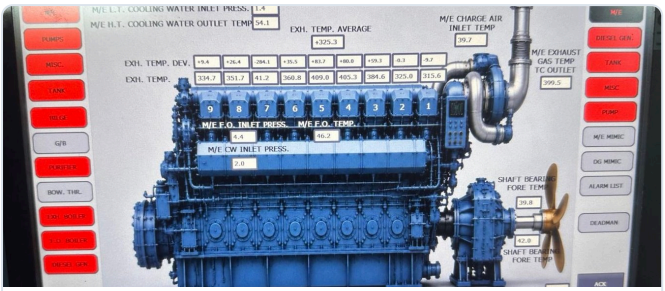
SCOPE
Main Engine + Auxiliaries
G/B, DG, pumps, tanks…



ECR SCADA workstation — main engine mimic (left) and generator DG1 alarm page (right). Example system screens.

EXAMPLE INTERFACES

1 System Screens



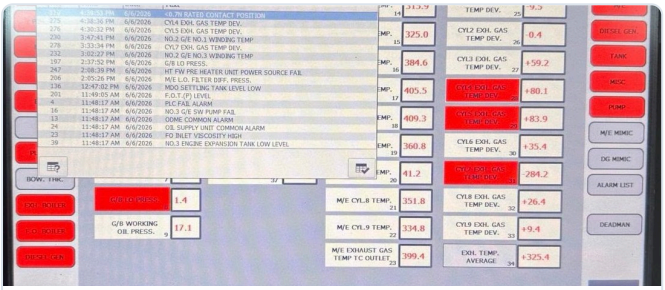
M/E Mimic

Exhaust gas temperatures for all 9 cylinders with deviation from average, T/C outlet, LO / FO / cooling water pressures and temperatures, shaft bearing and G/B values.



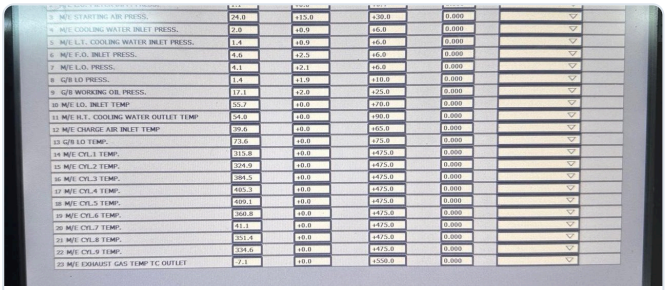
M/E Alarm Panel

Grouped digital alarm points, colour-coded active alarms, ACK / Horn Stop and system group navigation.



Active Alarm List

Date and time-stamped alarm log: exhaust deviations, winding temperatures, pressure and level alarms.



Analog Alarm Settings

Low / high limits, alarm delay (s) and alarm cut-off per channel — adjustable by authorised personnel.

Images show example system screens. Screen layouts are adapted to the vessel's I/O list and your requirements.

WHAT IS INCLUDED

2 Scope of Supply

Engineering & Design

Survey of existing system and I/O, I/O list, electrical and cabinet drawings, functional description, class approval documents.

PLC Hardware

Siemens SIMATIC S7 CPU, digital / analog I/O, PT100 / thermocouple modules, power supplies and UPS.

HMI Panels

SIMATIC touch HMI panel(s) — ECR, local and/or bridge (as requested).

ECR SCADA Station

Industrial PC, dual monitors, mimic pages, alarm log and history screens.

Cabinet Manufacturing

Marine-type PLC cabinet with terminals, relays, fuses, internal wiring and cabinet temperature monitoring.

Software

PLC & HMI programming; M/E, pumps, tanks, bilge, G/B, purifier, bow thruster, boiler and DG alarm groups; ACK, Horn Stop, Deadman.

Commissioning & Testing

Installation supervision, loop tests, FAT / SAT, support during class survey.

Training & Documentation

Crew training, as-built drawings, operating manual and program backups.

ESTIMATED BUDGET

3 Pricing

#	ITEM	ESTIMATED RANGE (USD)
01	Engineering, system design and documentation	5,000 – 8,000
02	PLC hardware (CPU, I/O modules, power supply, UPS)	12,000 – 20,000
03	HMI touch panel(s)	6,000 – 11,000
04	PLC cabinet manufacturing and internal wiring	6,000 – 10,000
05	ECR SCADA workstation (PC, monitors, software)	5,000 – 9,000
06	PLC & HMI software development	6,000 – 9,000
07	Installation supervision, commissioning and testing	4,000 – 6,000
08	Training and handover documentation	1,000 – 2,000
Total Estimated Budget		**45,000 – 75,000 USD** EXCLUDING VAT

Item ranges are indicative. Prices will vary according to your requirements; the final breakdown will be confirmed in the final quotation.

4 Not Included in the Price

- ✕ Any **sensors**, transmitters, temperature elements (PT100 / thermocouple), pressure and level switches or other field equipment that need replacing.
- ✕ Any **cables** that need replacing / renewing.
- ✕ **VAT** — prices exclude VAT.

5 What Determines the Price

- Total number of monitored I/O points (digital / analog)
- Number and location of HMI panels (ECR, bridge, local)
- Number of SCADA workstations and monitors
- Auxiliary systems to be integrated
- Redundancy requirements (CPU, network, power supply)
- Classification society requirements and approval
- Existing equipment to be reused
- Extras such as remote monitoring / data logging

6 Next Steps

01 Survey
On-board inspection of the existing system and preparation of the I/O list

02 Scope
Confirming the system scope together, according to your requirements

03 Final Quotation
Submission of firm price, delivery time and payment terms

Please do not hesitate to contact us with any questions. We look forward to hearing your preferred dates for the on-board survey.

Best regards,

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